4 Task 2: Proximal Policy Optimization

We continue PPO from the supplied midpoint (Qwen2.5-1.5B LoRA policy, 0.5B LoRA critic, frozen reward model r_ψ) for 20 updates (one prompt, two epochs, LR 3×10^{-6}, GAE γ=1, λ=0.95, ϵ=0.2, β_{KL}=0.1). Ablations fork 8-update runs from the identical state with common random numbers over $\epsilon \in \{0.05, 0.2, 0.5\}$ and $\beta_{\text{KL}} \in \{0, 0.1, 0.2\}$. All conditions use the same 200 held-out prompts with paired-bootstrap 95% CIs (Table 1).

Table 1: Training statistics (mean $\pm$ std over updates) and held-out reward change vs. the midpoint (r_ψ=1.533; 95% paired-bootstrap CI). †The three ϵ forks are bit-identical and share a row (equal to β=0.1).

Run	$\mathcal{L}_V$	Clip frac.	$\max\lvert\rho-1\rvert$	$\hat{D}_{\text{KL}}$ (10^{-4})	r_ψ (train)	Entropy	Length	Held-out Δr_ψ
Standard (20 upd.)	4.30 ± 1.45	0	0.078	-0.1 ± 5.1	1.19 ± 1.37	1.04 ± 0.51	241 ± 186	$+.034$ [$-.034, .106$]
$\epsilon \in \{.05, .2, .5\}$†	5.47 ± 1.11	0	0.031	-0.3 ± 6.7	0.86 ± 1.05	0.87 ± 0.43	224 ± 239	$+.002$ [$-.055, .068$]
$\beta_{\text{KL}} = 0$	5.59 ± 1.14	0	0.027	-1.2 ± 8.8	0.73 ± 1.18	0.86 ± 0.43	219 ± 243	$+.019$ [$-.072, .110$]
$\beta_{\text{KL}} = 0.2$	5.79 ± 1.11	0	0.029	$+2.6 \pm 5.5$	0.82 ± 1.14	0.84 ± 0.42	219 ± 243	$+.051$ [$-.040, .144$]

4.1 Baseline and Critic Dynamics

The surrogate loss is negligible ($|\mathcal{L}_\pi| \sim 10^{-4}$) because $\rho \approx 1$, so learning hinges on the critic, which fails. Explained variance is negative in 20/20 updates (median -5.3), and the mean value loss (4.30) exceeds the reward variance (2.16), so V_ϕ is worse than a constant predictor. With terminal rewards and γ=1, GAE accumulates $\delta_t = V_{t+1} - V_t$ inside each response, so an uncalibrated critic adds token-wise noise to every advantage, which whitening cannot remove. This shows up as an erratic policy-gradient norm (CV 0.78) and a critic gradient that exceeds its clip by 36–287$\times$ in every update (Fig. 3). One trajectory per update is too little data to fit a length-dependent value function.

4.2 RQ1: Clipping (ϵ)

The clip binds only when $|\rho - 1| > \epsilon$, but $\max|\rho - 1| = 0.031$ across all ϵ forks. The clip fraction is therefore 0 in every update, the clipped and unclipped gradients coincide, and the three forks are bit-identical in training and held-out metrics (Fig. 1). A cached-batch probe that seemed to show clipping (0.55/0.34/0.18) had used mismatched prompts. Corrected, it gives 0.108/0.0085/0.0056, equal to a noise-only floor (0.112/0.0084/0.0057) from bf16-versus-fp16 numerics. The reason is the step size: with one prompt and two epochs, $\rho \neq 1$ arises only after a single Adam step at LR 3×10^{-6}, which moves probabilities by at most 3%. PPO here is a policy gradient inside the trust region, and ϵ cannot affect stability, entropy or reward.

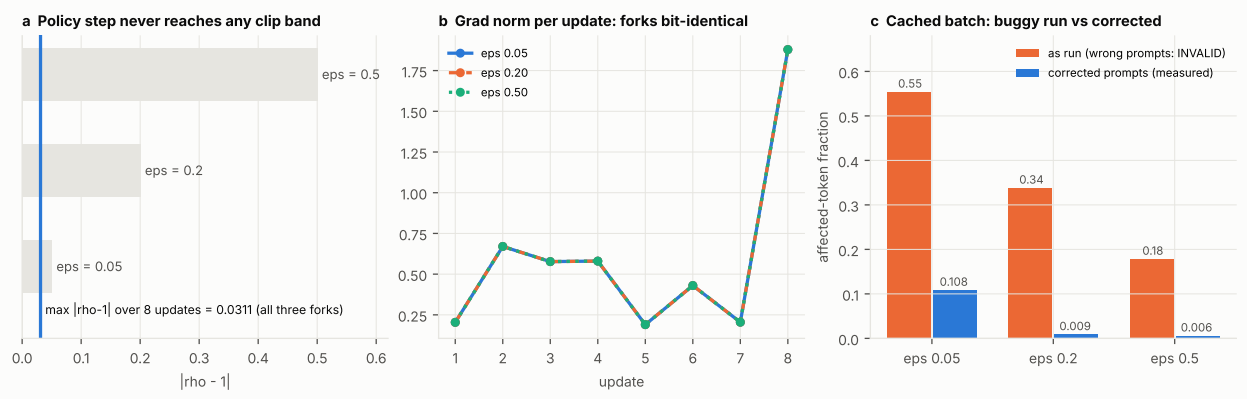


Figure 1: Clipping. (a) $\max|\rho - 1|$ (0.031) never reaches any band. (b) Gradient norms of the three ϵ forks coincide. (c) Cached-batch affected fraction before and after the prompt correction; the corrected value equals the numerical floor.

4.3 RQ2: KL Pressure (β_{KL})

Because update 1 is identical across forks, later differences are caused by β. KL and entropy diverge first (update 2, on identical sampled tokens), then reward and length (update 3). Fluency is unchanged (distinct-2 0.931–0.935 vs. 0.934). This order reflects measurement sensitivity rather than a cascade: log-probability statistics register 10^{-4} changes, while reward moves only when a sampling decision flips, an $O(1)$ bifurcation whose winner is arbitrary. The ordering is therefore not monotone in β (Fig. 2). The penalty never binds: $\beta|\mathrm{KL}_{\mathrm{seq}}| \leq 0.02$ (<2% of the reward s.d.), and $\nabla\mathrm{KL}$ vanishes at the reference. Even $\beta{=}0$ shows no drift or entropy collapse, so these runs cannot show that the penalty *prevents* collapse, only that it was inactive.

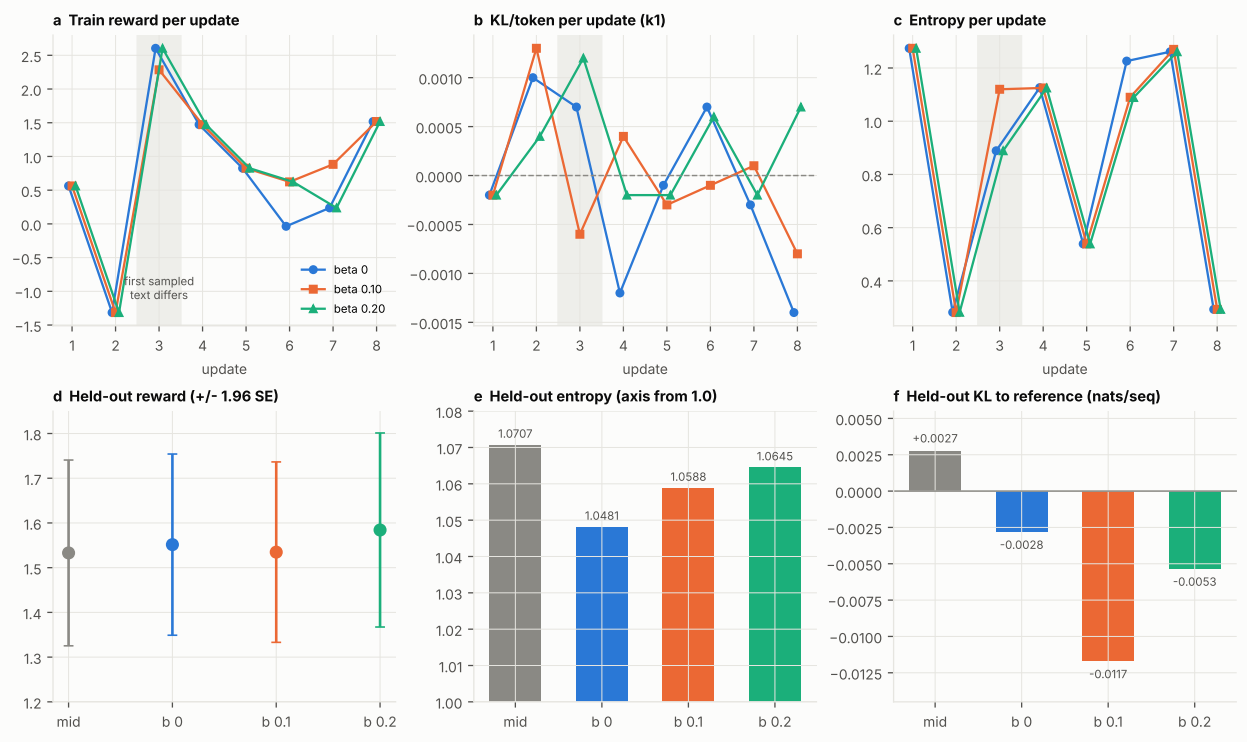


Figure 2: KL forks. (a–c) Training reward, KL and entropy; the grey band marks the first update whose sampled text differs. (d–f) Held-out reward (± 1.96 SE), entropy and KL; no condition separates from the midpoint.

4.4 RQ3: Reward Reliability and Over-optimization

No over-optimization is detectable because the policy never drifts. All held-out CIs include zero, $|\mathrm{KL}_{\mathrm{seq}}| \leq 0.012$ nats, and 63–67% of responses match the midpoint verbatim. The proxy is nonetheless fragile. It has 0.674 pairwise accuracy, a length correlation of 0.34, and a per-sample noise floor (0.126) larger than any mean effect. Right-truncating prompts at 256 tokens (17.5% of the evaluation set) removes the assistant header and lowers reward by 2.14 ($p<10^{-4}$; Fig. 4). Two same-seed pairs illustrate this. On idx 178, reward rises from -0.13 to $+1.11$ for a genuinely more helpful, though three times longer, answer. On a truncated translation prompt (idx 72), it rises from $+0.15$ to $+2.23$ for a looping, unterminated output (distinct-2 $0.69 \rightarrow 0.57$). Both are sampling bifurcations, not learned exploits, but they expose the biases (length, tolerated repetition, truncation) that stronger optimization would exploit first.

Limitations. The study uses a single seed and one prompt per update. The standard-run adapter used for the held-out evaluation was regenerated by retraining, so its identity with the original is assumed.

A Additional Task 2 figures

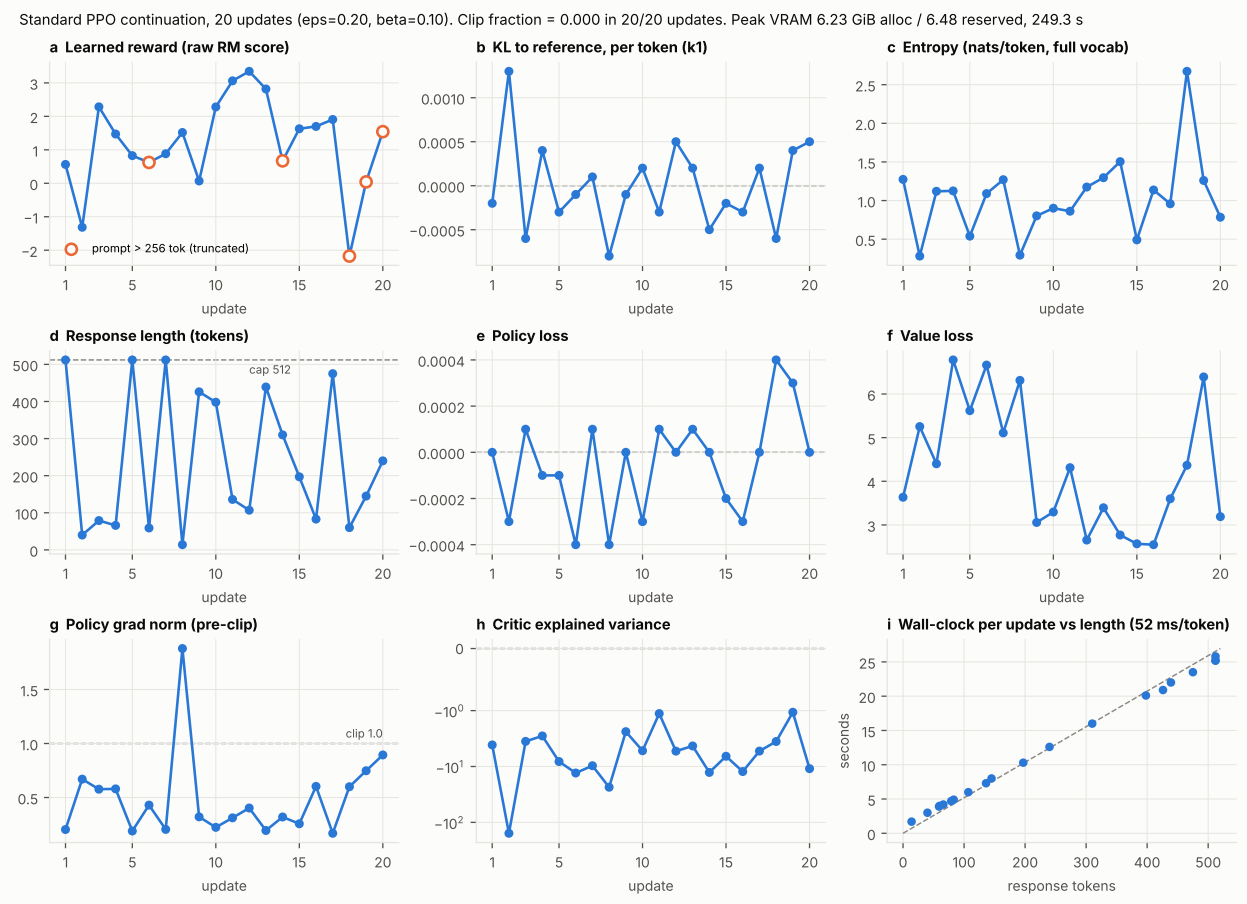


Figure 3: Standard run (20 updates): reward (hollow markers are truncated prompts), KL, entropy, length, policy and value loss, gradient norm, critic EV (negative throughout) and wall-clock. Clip fraction is 0 in every update.

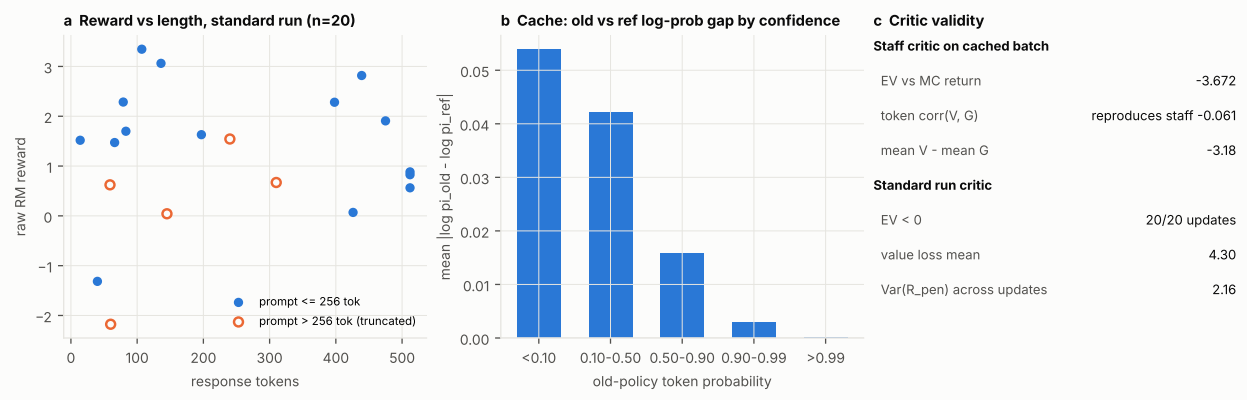


Figure 4: Reliability. (a) Reward versus length, standard run (hollow markers are truncated prompts). (b) Cached-batch numerical gap by token confidence. (c) Critic validity summary.